CST463 - Advanced Machine Learning

CST463 (Advanced Machine Learning) Syllabus

Summary

In this course you will learn to use advanced machine learning methods, including dimensionality reduction, ensemble methods, neural nets, and methods for working with time series data.

- **Grading:** *Details can be found in grading section.*
- **Late work:** *Details can be found in late policy.*
- **Getting help:** *Details can be found in Instructor Information.*

Course Information

Course Details

- **Course Title:** Advanced Machine Learning
- **Course Number:** CST463
- **Prerequisites:** CST383

Course Description

In this course you will learn to use advanced machine learning methods, including dimensionality reduction, ensemble methods, neural nets, and methods for working with time series data.

Course Objectives

At the end of this class, you should be able to: - Apply deep learning to machine-learning problems of moderate complexity, using Google's TensorFlow library and Python. - Explain and implement machine learning algorithms using loss functions and optimization, including the use of backpropagation in neural nets. - Explain the principles behind dimensionality reduction and ensemble methods (including random forests and boosting). - Apply machine learning to time series data sets.

Topics Covered

The major topics covered in class are: 1. Working with Data & Math 2. Gradient Descent 3. Neural Network Basics 4. TensorFlow 5. Vision Models 6. Time Series 7. Text Models

Details can be found in the course calendar

Personnel

- Dr. Sam Ogden (instructor)
  - E-mail: `sogden@csumb.edu`
  - Web: **`https://csumb.edu/scd/sogden/`**
  - Office: BIT205
  - Office Hours: 2pm-3pm Mondays & 10am-11pm Thursdays, or by appointment

Contacting me You can reach me via email (sogden@csumb.edu), or via slack by either DM or tagging me in a message to the channel. In general, I will not see or respond to canvas messages/comments so please use slack or email to contact me.

Materials

- Course textbook: (both available on O'Reilly Media for free)
  - Deep Learning with Python, second edition. Francois Chollet, Manning, 2021. (primary text)
  - Hands-on Machine Learning with Scikit-Learn, Keras, and Tensorflow, third edition. Aurélien Géron, O'Reilly Media, 2019.
- Course lecture recordings: Recording location
- Canvas Course Page: Canvas link
- Syllabus: CST463-syllabus.html

Grading

There are three groups of assignments, briefly described in the table below. **Getting below a 40% in any of these three groups will result in failing the class.**

Assignment Kind	Value
Programming assignments	36%
Exams	36%
Group Project	16%
Participation	12%

Programming Assignments

Programming assignments will consist of three parts: - Development of a library of functions that perform a set of tasks - Visualization and analysis using the functions developed during the first half - Reviews of your classmates work

The goal of these three parts is to give you experience in the three major aspects of data analysis and machine learning: development of new module and code, analysis of data, interpreting the work of others.

Assignments will generally span two weeks, with the first week focusing on the development of library functions and review of the previous week's analysis, and the second week will focus on your own analysis and visualization.

Therefore, there will be three parts to how you are assess: 1. The python code you develop will be checked against unit tests to ensure expected functionality (roughly 50%) 2. Your visualization and analysis, based on how well I and your peers understand your interpretation and the validity (roughly 30%) 3. Your analysis of your peer's visualization and analysis. (roughly 20%)

Exams

There will be three exams in this course worth each worth roughly 12% of your grade. Each one will cover a mixture of new and old material, with an emphasis on new material. During the exam you will be allowed to use a calculator but no other electronics, and have a single sheet of notes.

These exams are spaced roughly 5 weeks apart and can be seen on the course calendar.

Note: there is no final exam for this class, but we may have a project presentations period during this block.

Participation

In-class participation has three components:

1. (5%) Learning Logs: reflect on the material you learned each week and use as notes for studying
2. (5%) Attendance: Show up to class on time every day
3. (2%) Weekly Surveys: to help me understand how to improve class

Grade Assignment

Grades as assigned based on the standard ranges, which are outlined below.

Grade Range	Letter Grade
$[97, \infty)$	A+
$[93, 97)$	A
$[90, 93)$	A-
$[87, 90)$	B+
$[83, 87)$	B
$[80, 83)$	B-
$[77, 80)$	C+
$[73, 77)$	C
$[70, 73)$	C-
$[60, 70)$	D
$[0, 60)$	F

Course Policies

Attendance

Students are expected to arrive to class on time and to attend every class. Class attendance is one of the strongest factors for student performance. Attendance will be tracked either manually or with an attendance quiz. While no retakes of a quiz or attendance will be allowed, the lowest 2 will be dropped.

Late Policy

No late work will be accepted and no extensions will be granted.

The only exception to this is that the coding parts of homework can be turned in late, for a penalty. However, this will impact your ability to complete the visualization component so it is not recommended.

If you do fall behind in class, please set up a time to meet with me to discuss getting you back on track – my goal is to have you succeed in class and I want to find a way to make that happen.

Academic Honor Code

You may talk to other students or LLMs for:

- Clarification on topics covered in class
- Clarification of what *provided* code is doing
- Generating more examples
- Understanding compiler errors

You may *not* talk to other students or LLMs for:

- Code and assignment answers
- Exam questions and answers

The use of LLMs

You may talk to other students or LLMs (e.g. ChatGPT) for:

- Clarification on topics covered in class (e.g. paging, API usage, etc.)
- Clarification of what *provided* code is doing
- Generating more examples
- Understanding compiler errors

You may *not* talk to other students or LLMs for:

- Code and checkpoints answers – i.e. do not copy-and-paste code or answers from other students or LLMs
- Exam questions and answers

Details At the highest level, the use of LLMs in this class is ***prohibited*** *for programming assignments* and ***encouraged*** *for improving your own understanding of material.* Large Language Models (LLMs), such as ChatGPT and GPT4.0, are powerful language generation models. They can be used to produce code and summarize text, as well as answer clarifying questions. They can be a very effective tool in a programmer's toolbox if used appropriately. In this class, you should use them as a resource similar to stackoverflow – a good resource that should be critically considered since the code might be wrong, and not as effective as coming to talk to a TA or instructor.

You may use LLMs for asking clarifying questions and generating examples. I will demonstrate a number of these such questions in class but "how do I use XX command?" and "how does a free list in memory management work?" are examples. Additionally, asking for example problems (e.g. "can I have five examples of turnaround time calculation with FIFO and RR scheduling?") is an excellent use[^4]. Further, if you are confused about what a homework problem is asking or how it relates to operating systems at a larger scale you can ask them[^5].

You may *not* use LLMs for producing answers on programming assignments, quizzes or exams. Asking for an LLM's solution to a coding assignment is conceptually the same as asking another student to write you code for you, and will be treated as such. While you may ask for individual parts to help understand the C language better (e.g. "how do I write a do-while loop in C"), using it for more than ~2 lines of code is not allowed. As a rule of thumb, a prompt of "write me a function to do…" indicates that you are headed in a problematic direction.

You are encouraged to use LLMs like you would a TA or an instructor. You *may not* use LLMs for producing code for programming assignments, but *may* use it to learn how to better use the language and to clarify topics. They can clarify complex topics we learned in class, give you alternative explanations and can be incredibly helpful in understanding errors and problems. However, it is important to be able to identify *good* examples and *bad* examples of LLM output.

University Academic Integrity Policy

For complete academic integrity policies, please refer to the CSUMB Academic Integrity Policy.

University Policies and Resources

Enrollment and Registration

For information about requesting an incomplete or withdrawal from the course, please refer to CSUMB's Enrollment and Registration Policy.

For grade appeals, consult the Grade Appeal Policy.

Disability Services

If you have a disability that may require academic accommodations, please contact the Student Disability Resources office as soon as possible. All discussions will remain confidential. Students who have already received approval for accommodations from SDR should schedule a meeting with the instructor as early in the semester as possible.

Collection of Student Work

Student work may be collected and used for course assessment and improvement purposes. By enrolling in this course, you consent to the collection and analysis of your academic work for educational assessment. All student work will be handled confidentially and in accordance with FERPA regulations.

Syllabus Change Policy

This syllabus is subject to change at the instructor's discretion to enhance learning or accommodate unforeseen circumstances. Students will be notified of any substantive changes (such as changes to due dates, point values of assignments, or major policy modifications) via Canvas announcement and/or email at least one week in advance when possible.

The process for syllabus changes: 1. Instructor identifies need for change 2. Revised syllabus section is prepared 3. Students are notified via Canvas and email 4. Change takes effect after notification period

Students are responsible for staying current with any announced changes to the syllabus.